

Annamalai Arunachalam

CTO|CDO, Head of Technology & Applications | Principal Enterprise Architect

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A dedicated and results-oriented technology leader with over 25+ years of comprehensive experience in software development, solution architecture, product design, implementation, consulting, and IT management. Proven track record in spearheading digital transformation initiatives, leading cross-functional teams, and delivering cutting-edge solutions. Adept at aligning technology strategies with business goals to drive growth and operational excellence. Recently focused on Generative AI and Agentic AI systems, with hands-on expertise in large language models, RAG architectures, model context protocols, and multi-modal AI interfaces, applying these innovations to enterprise-grade cloud-native solutions.

Core Competencies

- **Innovation and Expertise:** Demonstrated deep expertise in data management, blockchain technologies, and generative AI—including large language models, agentic workflows, RAG pipelines, and model context protocols—driving innovation and enhancing business capabilities with cutting-edge, AI-enabled solutions.
- **Digital Transformation:** Extensive experience in spearheading digital transformation initiatives, including cloud computing, solution architecture, product design, implementation, system integration, and client services, with recent focus on AI-first enterprise platforms.
- **Strategic and Inspirational Leadership:** Strong background in strategic planning, budget management, contract negotiation and resource optimization, ensuring efficient operations and cost-effectiveness. Proven ability to inspire and motivate teams to achieve excellence and drive organizational success.
- **Vendor and Remote Management:** Proficient in managing vendors, remote development teams, and outsourced operations across global locations, ensuring seamless collaboration, delivery quality, and operational excellence.

Skillset

- **Blockchain:** Skilled in smart contract solution design and developing blockchain-powered data clean rooms using RUST, GO, Substrate, Hyperledger, Ethereum, Solidity, Node.js, and IPFS.
- **Generative AI and Agentic Systems:** Experienced in designing and deploying GenAI solutions using large language models (GPT, Claude, Gemini), CrewAI, Agno, MCP, Pydantic AI, LangChain, and vector databases. Built agentic AI prototypes leveraging RAG architecture, model context protocols, and multi-agent orchestration with both cloud-hosted and local models. Proficient in NL2SQL, prompt engineering, and document intelligence using similarity search and LLM chaining.
- **Cloud Computing:** Proficient in Azure, GCP, and AWS, with expertise in technology evaluation, cloud architecture, DevOps management, cost optimization, and managing vendor relationships.
- **Data Management & Analytics:** Competent in RDS, Athena, BigQuery, GCP Analytics Hub, Synapse Analytics, and DataFactory, delivering scalable analytics and data collaboration frameworks.

- **Full Stack Development:** Proficient in Python and its AI/ML ecosystem (Pandas, NumPy, Scikit-Learn, SciPy, TensorFlow/keras), extensively used for researching, evaluating, and developing Generative AI prototypes. Possess working knowledge of GO, Rust, ReactJS, Node.js and SQL— applied primarily in the context of architecting cloud, data, and blockchain components for a federated, privacy-compliant multi-party data collaboration platform.

Experience

Saptharushi

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Jul 2021 - Present

- **Data Clean Room (ATOM):** Designed and delivered a **privacy-preserving**, decentralized data clean room platform (ATOM) for MarTech and AdTech, enabling **secure multi-party data collaboration**. Employed a containerized microservices architecture integrated with message queues and blockchain protocols for verifiable data collaboration across hybrid cloud environments for enhanced audience targeting for digital advertising.
- **Generative AI Solutions:** Developed a natural language querying mechanism (**NL2SQL interface**) integrated with a distributed query engine, enabling users to input natural language queries and receive SQL-generated results. Implemented **RAG functionality** to derive insights from client documents through vector DB similarity searches.
- **Agentic AI Prototyping & RAG Applications:** Designed and developed multiple **Agentic AI** applications using both local and hosted LLMs, orchestrating autonomous task planning and execution workflows. Integrated tool use through **MCP** that enables the agents to avail data services when needed. Integrated Retrieval-Augmented Generation (RAG) pipelines with semantic search, enabling dynamic knowledge retrieval from client datasets.
- **Consent Management System:** Designed a **blockchain-based consent management solution**, embedding digital wallet capabilities to ensure user agency, consent traceability, and compliance with global privacy regulations (e.g., GDPR, PDPA).
- **Comprehensive IT and Resource Management:** Managed IT infrastructure, vendor relationships, and budgets, achieving significant cost savings and efficiency improvements. Oversaw the delivery of the flagship product 'ATOM' within budget, saving SGD 2M through strategic resource optimization.
- **Leadership and Innovation:** Led **R&D initiatives**, provided **mentorship** to technical teams, maintained thorough technical documentation, and architected solution, cloud, data, generative AI, and blockchain architectures, driving product development, deployment, and delivery.

Sembcorp Industries

Digital Applications Lead

Oct 2019 - Jun 2021

- **CTRM/ETRM Implementation:** Conducted business analysis to identify key pain points in trading and risk management, leading to the deployment of CTRM products that improved efficiency by 25% and reduced manpower costs by 50%.
- **Trade Configuration and Management:** Configured physical trades with inventory capability and logistics management, ensuring accurate accounting transactions and compliance with company standards.

- **Risk Management and Reporting:** Managed physical and paper transactions, exporting data to the Lacima system for risk evaluation using parametric VAR and Monte Carlo simulation, resulting in comprehensive market exposure reports.
- **System Integration and UI Customization:** Integrated accounts with SAP via scheduled batch transfers, customized the user interface with real-time validations and alerts, and configured MFA and authorization for secure user access.
- **Team Leadership and Cost Management:** Hired, mentored, and managed support staff, led data-related functions and system integrations, and achieved SGD 200k in cost savings through strategic resource allocation and outsourcing, increasing project profitability by 15%.

Mitsubishi

Enterprise Solutions Architect

Apr 2015 – Sep 2019

- **CTRM/ETRM Deployment:** Headed the successful implementation of the CTRM/ETRM system Aspect, enhancing trade processing speed by 20%. Evaluated solutions, negotiated costs, planned the program, and delivered in phases, while managing budgets, remote resources, and vendor contracts, reducing costs by 50%.
- **Risk and Reporting:** Developed and customized VaR modelling, MtM, PnL attribution, position exposure, credit exposure, and reconciliation reports. Successfully resolved major issues in the VaR and Credit Modules, implementing cost-effective solutions.
- **Automation and Integration:** Implemented RPA with NTT Office Robot for price data processing, automated SAP journal entries, and designed a comprehensive Aspect system including FrontOffice, MiddleOffice, and BackOffice functionalities.
- **Technology and IT Management:** Managed IT outsourcing, contract negotiations, budget preparation, technology evaluation, and IT procurement. Successfully migrated on-prem servers to Azure, reducing maintenance costs by 30% and increasing system reliability by 25%.
- **Leadership and Collaboration:** Conducted demos, analyzed business requirements, negotiated vendor contracts, and collaborated with senior management on IT strategy. Effectively managed a diverse team across time zones, fostering a cohesive working environment.

Aspect Enterprise Solutions

Pre-sales Solutions Director

Oct 2010 - Mar 2015

- **CTRM/ETRM Implementation:** Took on the position of Presales Solutions Director at Aspect, managing solution proposals, RFPs, RFQs, contract evaluations, account management, sales pitches, client presentations, demos, POCs, and pilot programs.
- **Client Engagement and Contract Negotiation:** Conducted scoping workshops, developed proofs of concept and RFPs, negotiated contracts, and supported clients during program rollouts. Successfully closed 6 new clients and renewed SaaS contracts for 4 existing clients.
- **Project Management and Implementation:** Led implementation and project management, designing and implementing interfaces, business processes, and approval workflows using Aspect's proprietary scripting tool. Developed custom queries and reports.
- **Product Development:** Architected and developed the Forward Curve Builder, enabling users to create forward curves based on existing price curves. Defined product roadmaps and coordinated with development teams to launch new modules for inventory and logistics management, trade templates, and FC builder.

- **Customer Success and Revenue Protection:** Successfully implemented CTRM solutions for over 8 customers, including complex projects for Chemoil, Concord Energy, Hanwa, Trafigura, MRI, and Itochu. Resolved product issues, resulting in the renewal of a significant SaaS contract, preserving a revenue stream of USD 3M.

OpenLink | Senior Technical Consultant | Jan 2009 - Sep 2010

Allegro | Senior ETRM Implementation Consultant | May 2005 - Dec 2008

Energy Market Company | Technical Consultant | Feb 2003 – Apr 2005

ISI-Dentsu | Senior System Analyst | Jan 1998 – Jan 2003

Education

- Curtin University of Technology | **MBA** | IT Strategic Management
- University of Madras | **B.Eng** | Computer Science & Engineering

Courses and Certifications

- NUS Professional Certification - LLMs, Advanced NLP and Generative AI.
- Coursera Certification - Blockchain Technologies by INSEAD.
- Microsoft Certification – Azure Solutions Architect Expert, Azure Data Engineer Associate.
- Microsoft Certification – Azure Fundamentals, Power Platform Fundamentals, Azure Data Fundamentals, Azure AI Fundamentals.
- GCP Certification– Digital Transformation, Innovating with Data, Cloud Identity, Architecting with GKE, BQ, Big Data and ML Fundamentals.
- AWS Certification - AWS Cloud Practitioner, AWS Well-Architect-Framework.
- Oracle DBA I and II, PL/SQL, Java, Weblogic and SonicMQ.
- Python, Sybase, PowerBuilder & VB.NET, ColdFusion.
- Allegro, Endur / Findur, Aspect, Lacima RiskEngine, Entrade.